2023 SSMO Tiebreaker Round Solutions

SMO Team

Tiebreaker Round Problem 1: From the phrase "Summer Solstice" (treating uppercase, lowercase, and the space as distinct, and using each character at most as many times as it appears), how many 4-character strings can be formed such that the first character is a capital letter and the second and third are not spaces?

Answer: $\boxed{1162}$

Solution: Firstly, note that the first character must be "S". Now, the remaining three characters are chosen from the following 12 characters

$$\{\text{S, u, m, e, r, o, l, s, t, i, c, space}\},$$

without replacement, except m and e may each be used twice. If all three remaining characters are distinct, then we have $11 \cdot 10 \cdot 10 = 1100$ valid arrangements, since the second and third characters cannot be spaces. Now, suppose one of the letters m or e is repeated. The remaining letter can either be one of 10 characters or a space. If the remaining character is a letter, there are 3 arrangements of the three characters. If it is a space, then there is only 1 arrangement (since the space cannot occupy the second or third positions). Thus, each repeated letter contributes $3 \cdot 10 + 1 = 31$ arrangements. Therefore, the total number of 4-character strings is

$$1100 + 2 \cdot 31 = \boxed{1162}.$$

Tiebreaker Round Problem 2: Let $P(x) = x^3 + 3mx^2 + 3nx + (m+n)$ be a real polynomial with positive real roots p, q, r (not necessarily distinct). Suppose that

$$(p+1)^3 + (q+1)^3 + (r+1)^3 + 3P(-1) = 0.$$

The value of $P(1)$ can be expressed as $a + b\sqrt{c}$, where a, b, and c are integers and c is not divisible by the square of any prime. Find $a + b + c$.

Answer: $\boxed{10}$

Solution: From Vieta's formulas, we have

$$p + q + r = -3m \quad \text{and} \quad pq + pr + qr = 3n.$$

So,

$$p^2 + q^2 + r^2 = (p+q+r)^2 - 2(pq + pr + qr) = 9m^2 - 6n.$$

Now, $P(x) = 0$ can be arranged as

$$x^3 = -3mx^2 - 3nx - (m+n).$$

We have

$$\begin{aligned}
&(p+1)^3+(q+1)^3+(r+1)^3+3P(-1)\\
=&(p^3+q^3+r^3)+3(p^2+q^2+r^2)+3(p+q+r)+3+3P(-1)\\
=&(3-3m)(p^2+q^2+r^2)+(3-3n)(p+q+r)+(3-3(m+n))+3P(-1)\\
=&(3-3m)(9m^2-6n)+(3-3n)(-3m)+(3-3(m+n))+3(-1+3m-3n+(m+n))\\
=&-27m^3+27m^2+27mn-27n\\
=&27(m-1)(n-m^2).
\end{aligned}$$

So, either $m=1$ or $n=m^2$. If $m=1$, then $p+q+r=-3<0$, a contradiction. That means we must have $n=m^2$. Substituting, we have

$$P(x)=x^3+3mx^2+3m^2x+(m^2+m)=(x+m)^3-(m^3-m^2-m).$$

The graph of $P(x)$ is the graph of x^3 shifted left m units and down m^3-m^2-m units. In order for $P(x)$ to have three real roots, it must have a triple root at $-m$, where $m<0$, meaning

$$m^3-m^2-m=0 \implies m(m^2-m-1)=0 \implies m=\frac{1-\sqrt{5}}{2}.$$

So,

$$P(1)=1+(3m)+(3m^2)+(m^2+m)=9-4\sqrt{5} \implies 9-4+5=\boxed{10}.$$

Tiebreaker Round Problem 3: For $n\geq 4$, let a_n be the maximum possible value of $P(n+1)$ given that $P(x)$ is a monic polynomial of degree n that satisfies $P(i)\in\{1,2,3,\ldots,n\}$ for $1\leq i\leq n$. The value of $\sum_{n=4}^{\infty}\frac{a_n-n!}{3^n}$ can be expressed as $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p+q$.

Answer: $\boxed{5}$

Solution: Note that the value of $P(n+1)$ is unique given $P(1),P(2),\ldots,P(n)$. From Lagrange interpolation, we have

$$P(n+1)=\sum_{i=1}^{n}c_iP(i)+n!,\quad c_i=(-1)^{n-i}\binom{n}{i-1}.$$

The value of $P(n+1)$ is maximized by choosing $P(i)=n$ when $c_i\geq 0$ and $P(i)=1$ when $c_i<0$. So,

$$a_n=n\left(\binom{n}{n-1}+\binom{n}{n-3}+\cdots\right)-\left(\binom{n}{n-2}+\binom{n}{n-4}+\cdots\right)+n!.$$

From the Binomial Theorem, we have

$$\begin{aligned}
&\binom{n}{n-1}+\binom{n}{n-3}+\cdots=2^{n-1}\quad\text{and}\\
&\binom{n}{n-2}+\binom{n}{n-4}+\cdots=\left(\binom{n}{n}+\binom{n}{n-2}+\cdots\right)-\binom{n}{n}=2^{n-1}-1.
\end{aligned}$$

So,

$$\frac{a_n-n!}{3^n}=\frac{\left(n\cdot 2^{n-1}-\left(2^{n-1}-1\right)+n!\right)-n!}{3^n}=\frac{(n-1)\cdot 2^{n-1}+1}{3^n}.$$

We have

$$\sum_{n=4}^{\infty} \frac{a_n - n!}{3^n} = \sum_{n=4}^{\infty} \frac{(n-1)\cdot 2^{n-1} + 1}{3^n}$$
$$= \sum_{n=4}^{\infty} \frac{(n-1)\cdot 2^{n-1}}{3^n} + \sum_{n=4}^{\infty} \frac{1}{3^n}.$$

Setting

$$S = \sum_{n=4}^{\infty} \frac{(n-1)\cdot 2^{n-1}}{3^n} = 3\cdot\frac{2^3}{3^4} + 4\cdot\frac{2^4}{3^5} + 5\cdot\frac{2^5}{3^6} + \cdots,$$

we have

$$\frac{2}{3}S = 3\cdot\frac{2^4}{3^5} + 4\cdot\frac{2^5}{3^6} + \cdots.$$

Subtracting the two, we get

$$\frac{1}{3}S = 3\cdot\frac{2^3}{3^4} + \left(\frac{2^4}{3^5} + \frac{2^5}{3^6} + \cdots\right)$$
$$= \frac{8}{27} + \frac{\frac{2^4}{3^5}}{1-\frac{2}{3}} = \frac{40}{81} \implies$$
$$S = \frac{40}{27}.$$

So,

$$\sum_{n=4}^{\infty} \frac{a_n - n!}{3^n} = \frac{40}{27} + \sum_{n=4}^{\infty} \frac{1}{3^n} = \frac{40}{27} + \frac{\frac{1}{81}}{1-\frac{1}{3}} = \frac{3}{2} \implies 3+2 = \boxed{5}.$$